

Advanced Analytics

Content

- Need for Advanced Analytics
- Advanced Analytics- Definition
- Business Intelligence vs Advanced Analytics
- Advanced Analytics Approach
- Tools and Technologies
- Data Driven Decision Making Process
- Scope of the Module
 - Descriptive, Predictive, Prescriptive
- Summary
- References

Applications : Then

OrderForm

Sales Order Number: SO00000000 Rev. 0

Customer: AW00000053

☐ Account Number:

☐ P.O. Number:

Billing Address: 99 Edgewater Drive, Norwood, 2062

Shipping Address: 99 Edgewater Drive, Norwood, 2062

Shipping Method: CARGO TRANSPORT 5

☒ Sales Territory: Northeast

Order Date: Ju

Due Date: Ju

Ship Date: Ju

Order Status: In pro

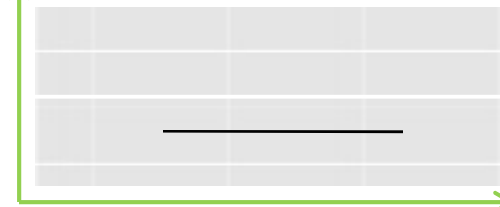
☐ Credit Card:

☐ Approval:

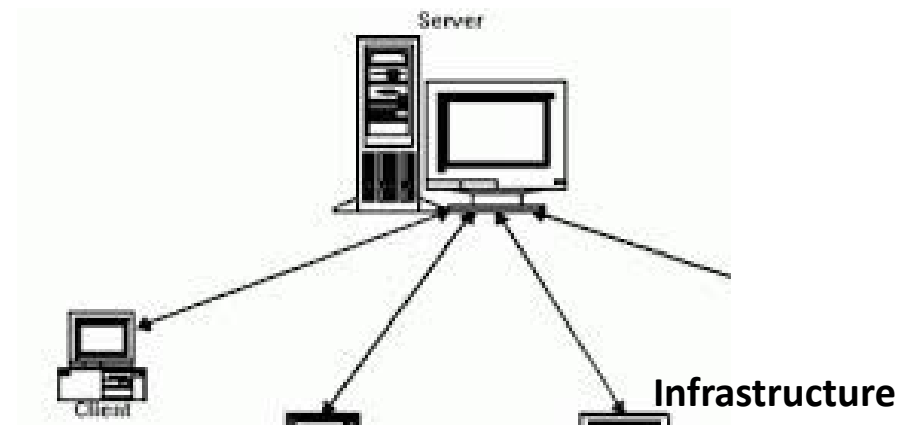
	Product Number	Product Name	Qty	Price	Discount	Disco %
	BK-M188-40	Mountain-500 Black, 40	2	149.99	No Discount	0.00%
	BK-R198-44	Road-750 Black, 44	4	99.99	No Discount	0.00%
	BC-R205	Road Bottle Cage	4	9.99	No Discount	0.00%
	HL-U509	Sport-100 Helmet, Black	6	24.99	No Discount	0.00%

Business Process Automation

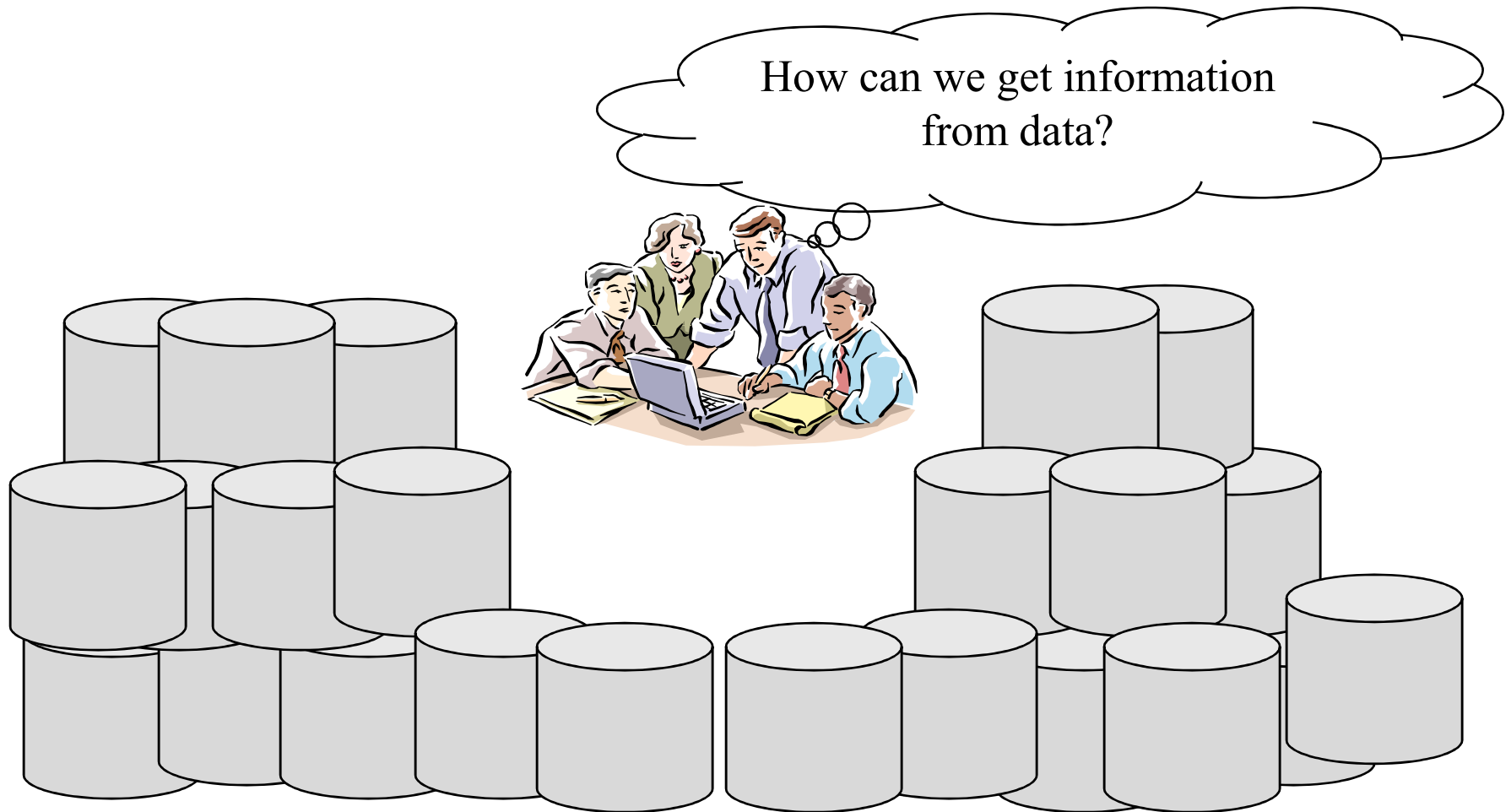
No of users
(thousands)



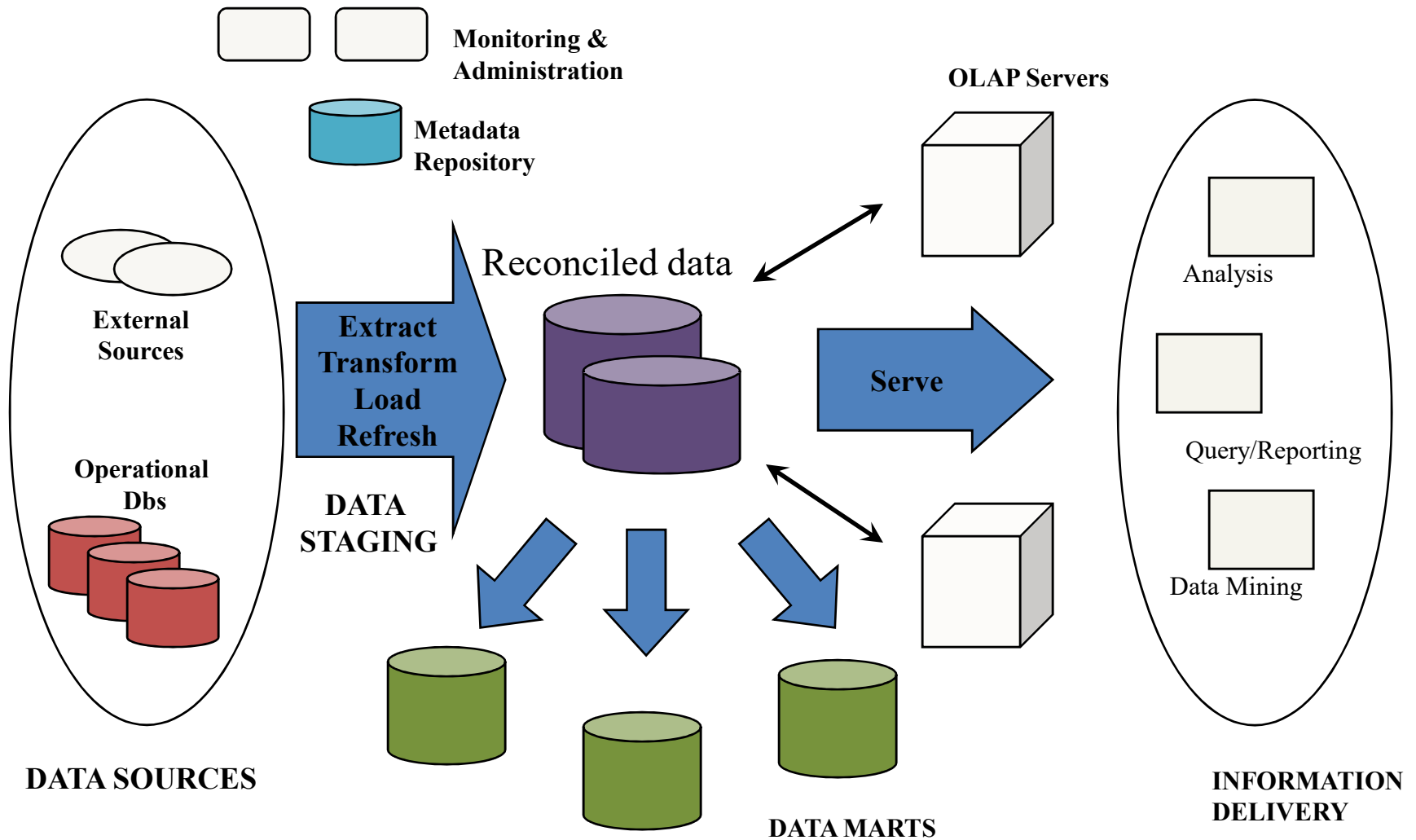
Time



1990s Scenario

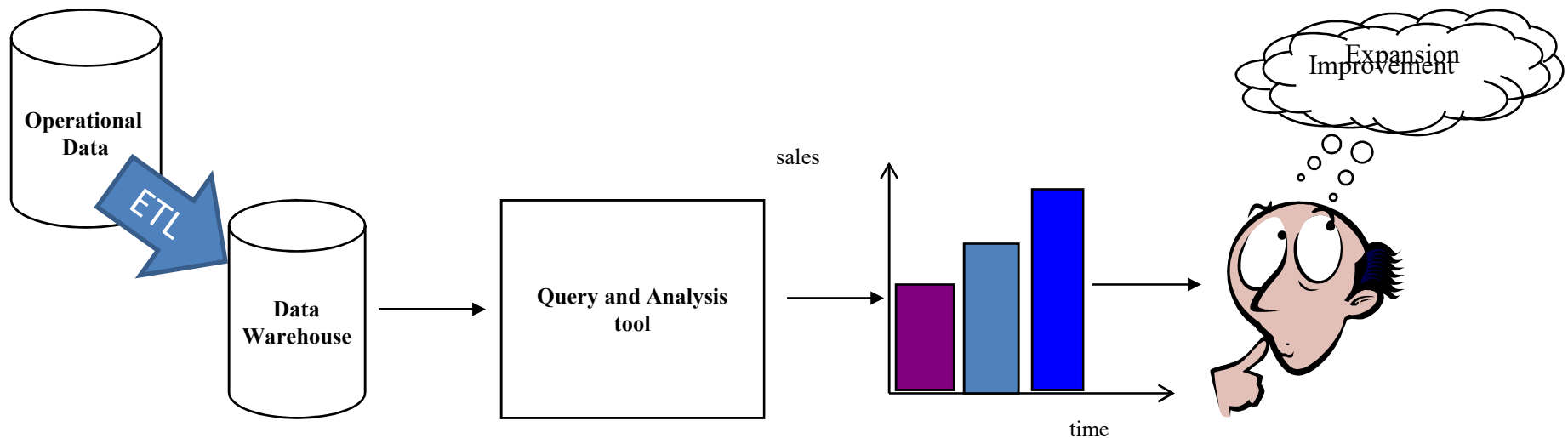


Business Intelligence

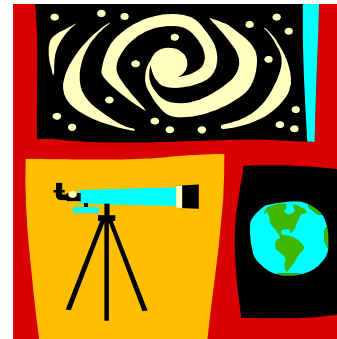
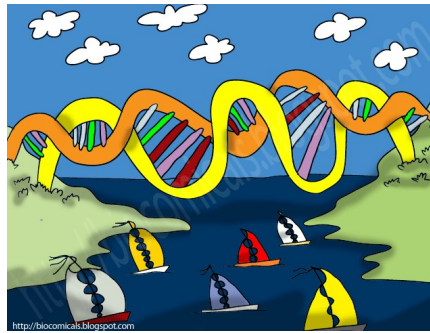


Data Analytics Use Case

Cakes & Cookies is a foods and beverages company. CEO wants that his company should grow. For this he need to take strategic decisions.



2000s Scenario

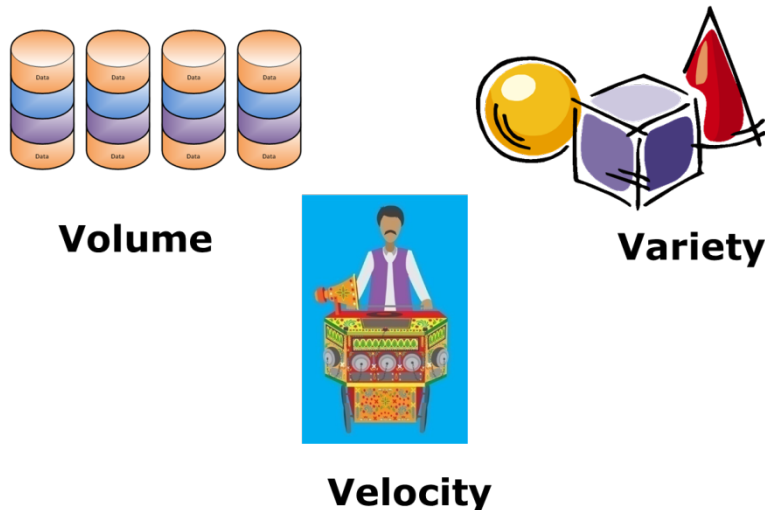


The sciences like astronomy and genomics who first experienced the data explosion in the 2000s, coined the term “Big Data.”

However, there is no single definition...

Big Data Definitions

Meta Group Report

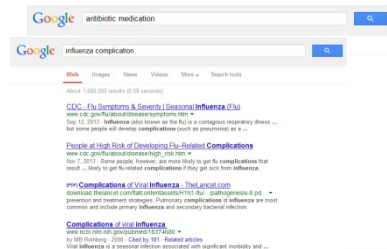


+ veracity, value, variability and more

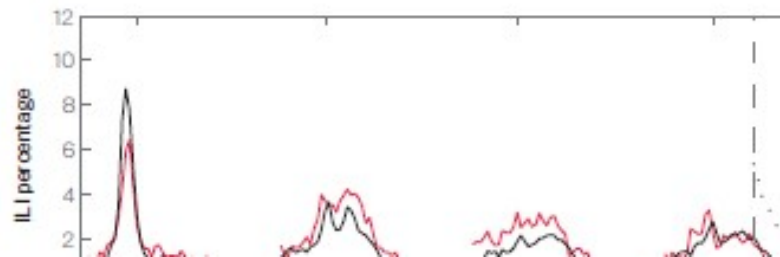
Gartner's Definition

"Big data is high-volume, velocity and variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."

Winter Flu Prediction



Mathematical Models



Applications : Now

Click to **LOOK INSIDE!**

The Data Warehouse Toolkit: The Complete Guide to Dimensional Modeling
 Ralph Kimball (Author), Margy Ross (Author)
 ★★★★★ (55 customer reviews)

Frequently Bought Together

Price for all three: \$124.08
 Add all three to Cart Add all three to Wish List
 Some of these items ship sooner than the others. Show details

- ✓ **This item:** The Data Warehouse Toolkit: The Complete Guide to Dimensional Modeling by Ralph Kimball Paperback \$53.30
- ✓ The Data WarehouseETL Toolkit: Practical Techniques for Extracting, Cleaning, Conforming, and Delivering by Ralph Kimball Paperback \$33.90
- ✓ The Data Warehouse Lifecycle Toolkit by Ralph Kimball Paperback \$36.88

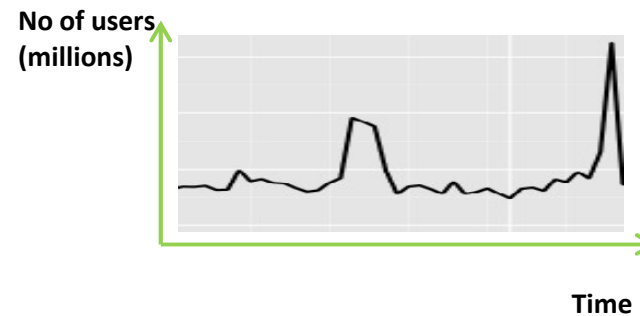
Customers Who Bought This Item Also Bought

Ralph Kimball
Margy Ross

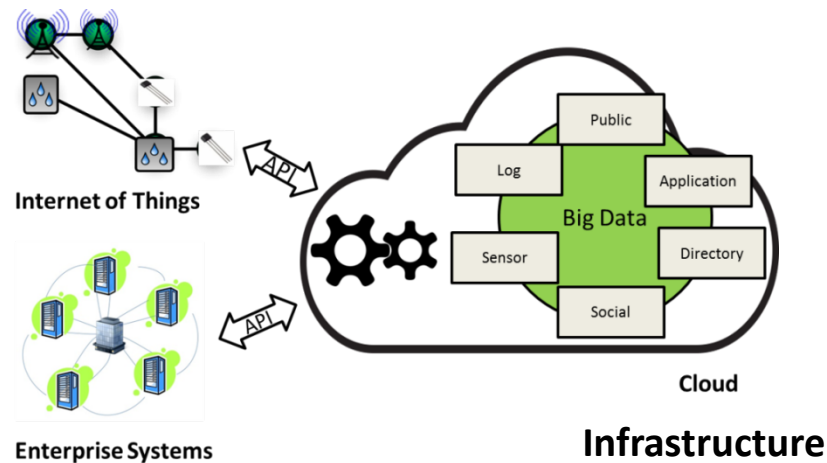
The Data Warehouse ETL Toolkit

The Data Warehouse Lifecycle Toolkit

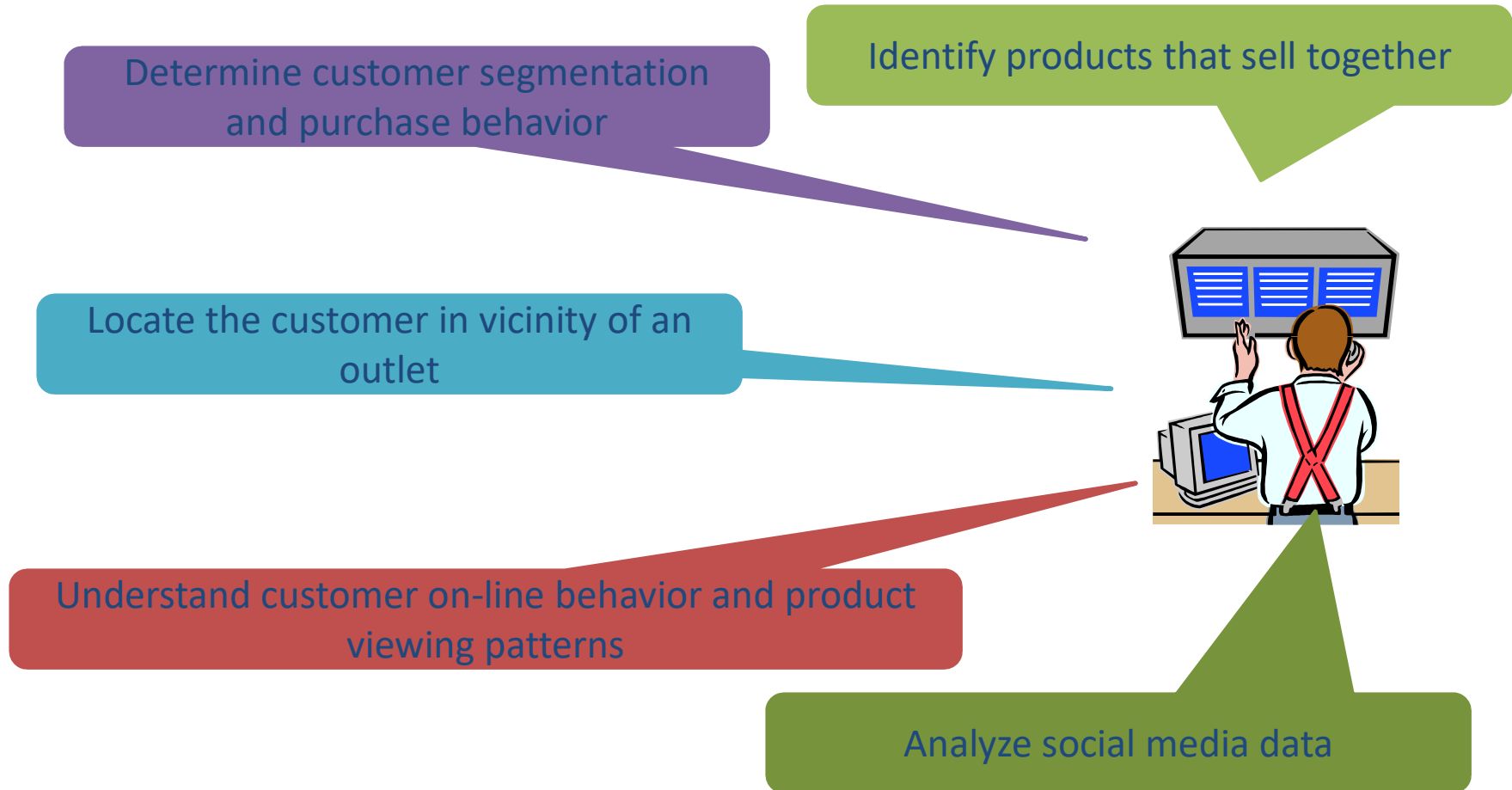
The Microsoft Data Warehouse Toolkit



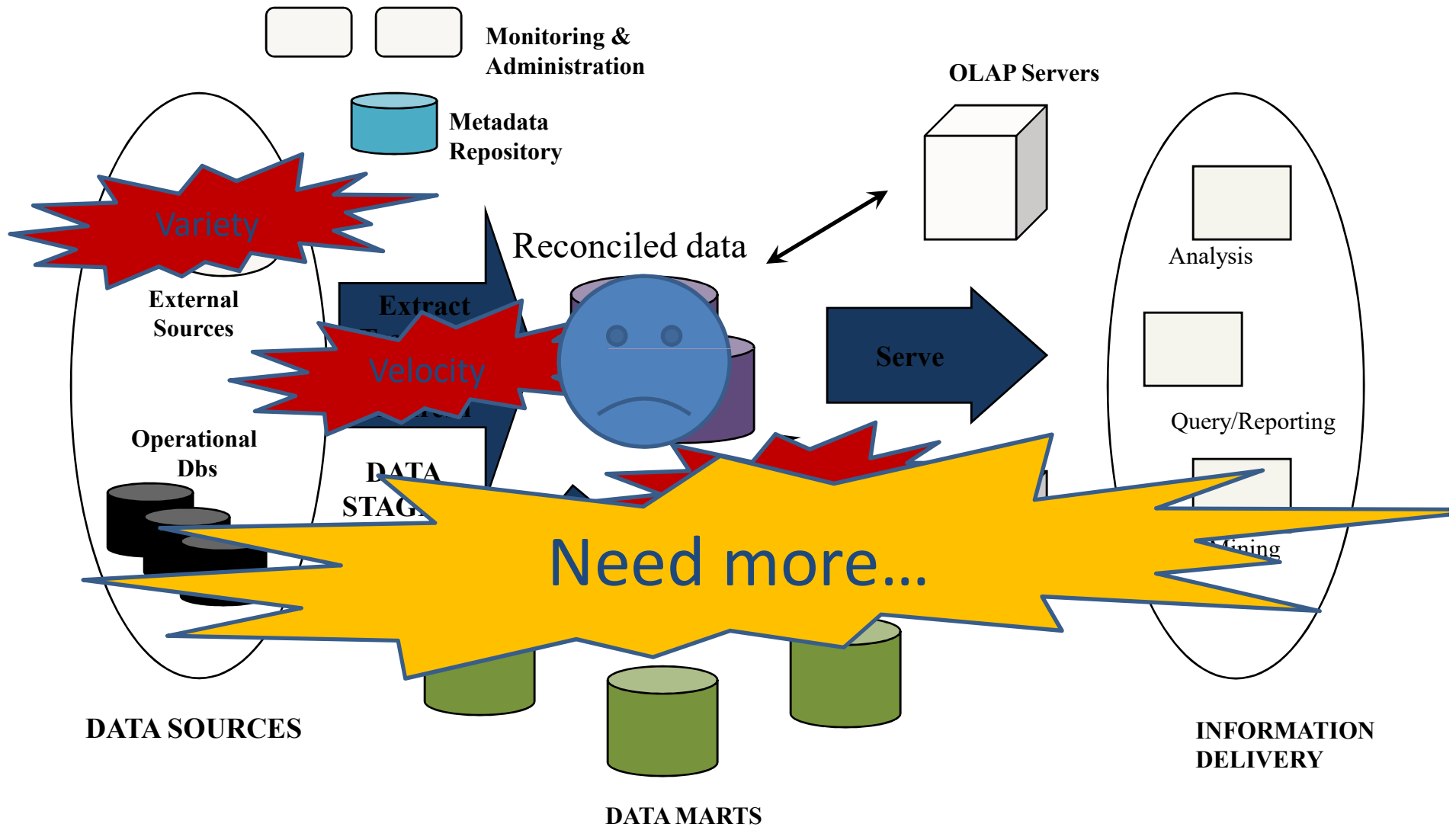
Business Process Innovation



Analytical Complexity



Business Intelligence



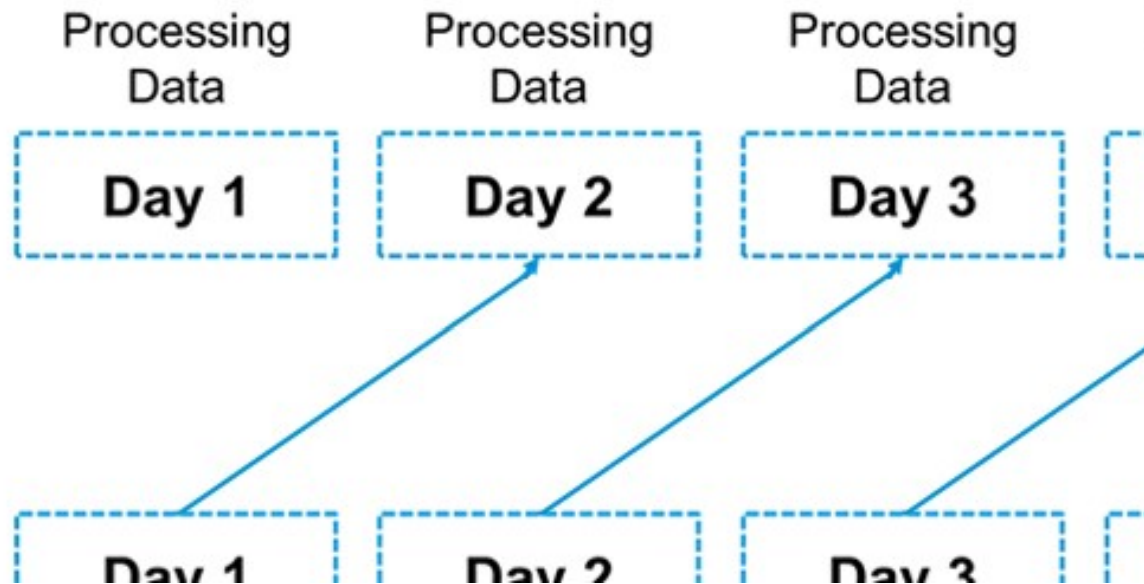
Big Data Analytics

- Big Data Analytics enables analysis of new types of data, rapidly increasing data volumes and flurry of data streams that may be left untapped by conventional business intelligence (BI) programs.
- It doesn't change traditional Data Warehousing or Business Intelligence architectures. But simply supplements them with new technologies and access methods.
- Two types
 - Batch Processing – Hadoop, Map Reduce
 - Real Time Processing – Spark Streaming

Big Data Storage Evolution



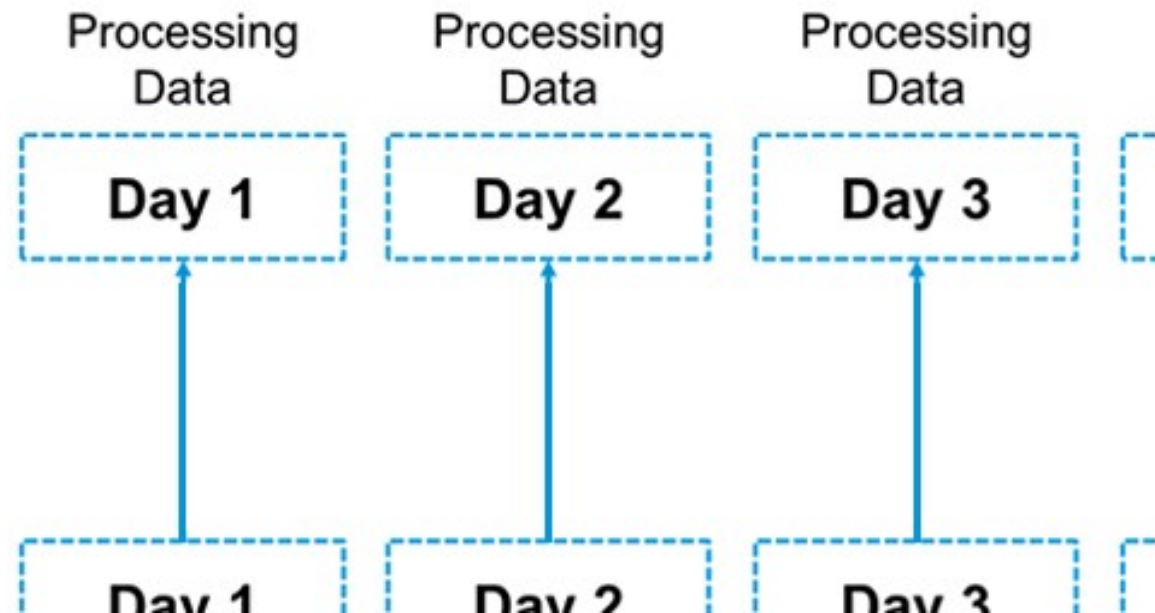
Batch Processing



Collect data over a period of time, input the batch into the system, process it and write the output.

- Hadoop processing data using Map Reduce

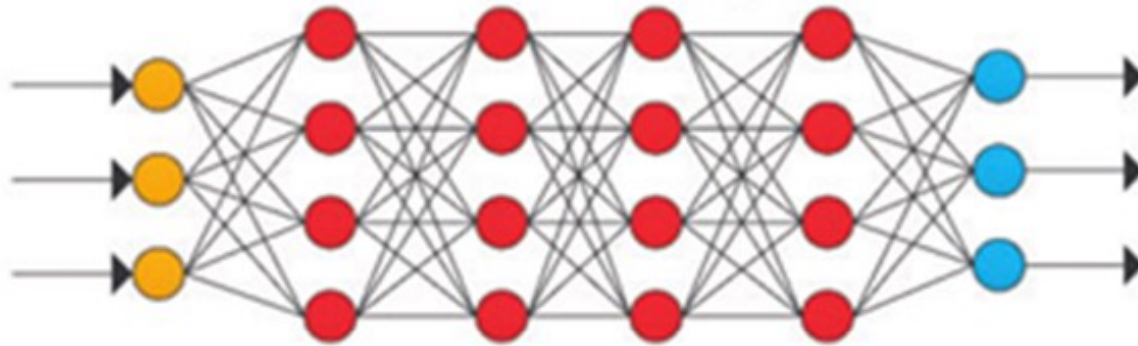
Real Time Processing



Continually input, process and output data. Data must be processed in a small time period.

- Real time processing in Spark, Real Time Dashboard

Resurgence of Neural Networks



“Deep Learning is a new area of Machine Learning research, which has been introduced with the objective of moving Machine Learning closer to one of its original goals: Artificial Intelligence. Deep Learning is about learning multiple levels of representation and abstraction that help to make sense of data such as images, sound, and text.”

Advanced Analytics

Gartner's Definition:

Advanced Analytics is the autonomous or semi-autonomous examination of data or content using sophisticated techniques and tools, typically beyond those of traditional business intelligence (BI), to discover deeper insights, make predictions, or generate recommendations.

Tools and Techniques

- Text mining
- Machine learning
- Pattern Matching
- Forecasting,
- Visualization
- Semantic analysis
- Sentiment analysis
- Neural networks
- Network and cluster analysis
- Multivariate statistics
- Graph analysis
- Simulation
- Complex event processing
- Optimization

Mission, Vision and Goals

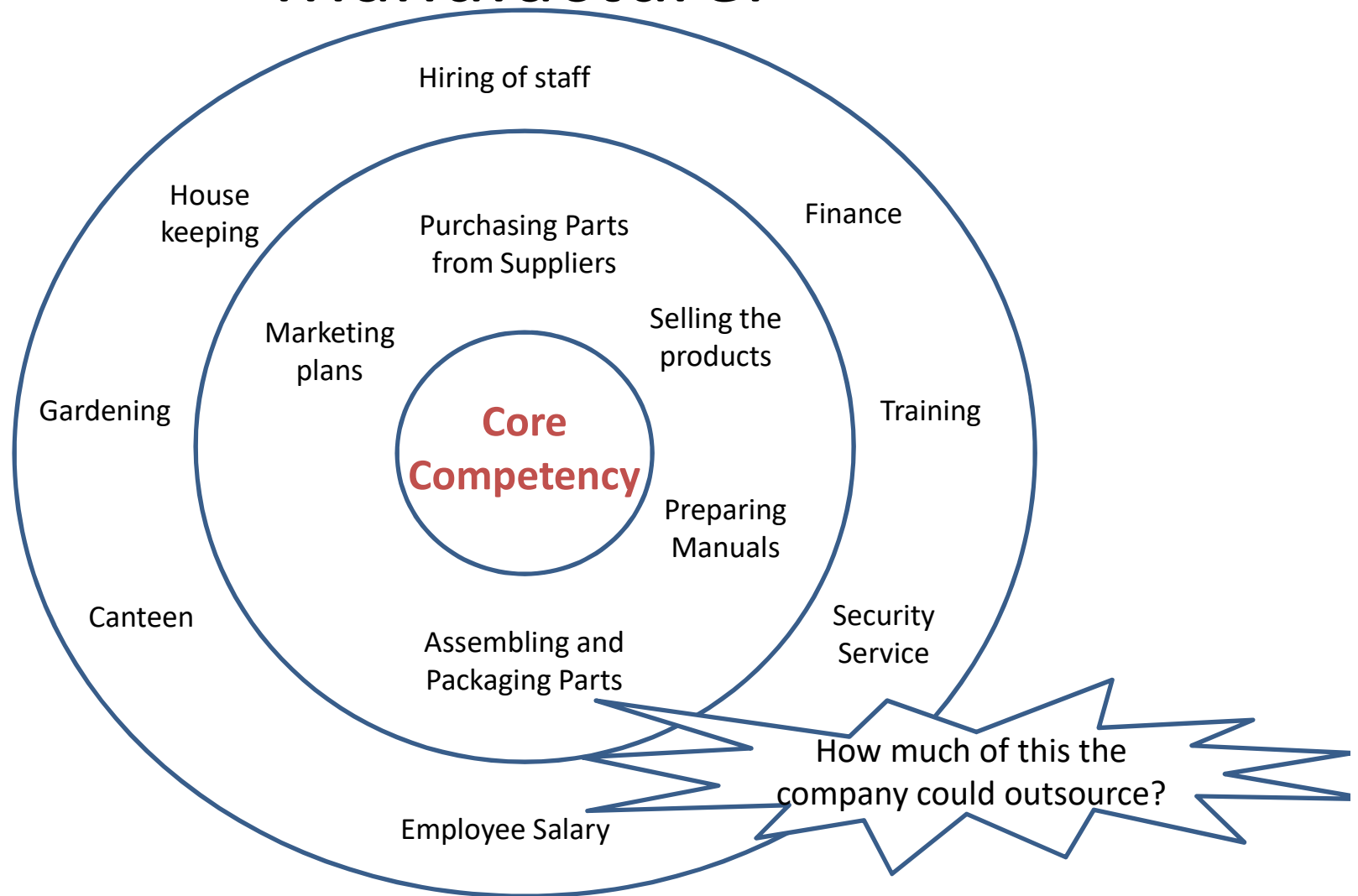
- A mission reflects the organization's past and present by stating why the organization exists and what role it plays in society.
- A vision states what the organization aspires to become in the future.
- Goals are the more specific aims that organizations pursue to reach their visions and missions.

Business Process

- A business process is an activity or set of activities that will accomplish a specific organizational goal.
- Primary processes
 - Purchasing raw material
 - Manufacturing the products
- Secondary processes
 - Repairing machinery
 - Cleaning, employing and training staff and so on
- Business Analytics should be about which business process to be supported as well as identify how added value is achieved.



Case Study: Home Computer Manufacturer



Core Competencies

- Not physical world: machinery , warehouse , distribution
- A company's core competencies lie in the field of knowing
 - how to handle internal processes
 - What customers want now and in future
- These competencies include
 - things the company is especially good at
 - which can secure its survival in the futureprovided it is capable of developing these so that they continue to meet the requirements in the marketplace.

Purpose

Business Intelligence

- Reports: gather and aggregate data to inform business users of the past performance of business operations.
- Lag information: to monitor, maintain and optimize existing processes.
- What was our revenue and profit last quarter?

Advanced Analytics

- Customer Analytics in a timely manner.
- Identify the root causes of success or failure in business operations
- Lead information: to revolutionize processes landscape.
- What is the risk of losing money in a new business venture?

Case Study: Euro Disney

- To avoid overcrowding of visitors in the amusement parks, and
- To optimize the distribution of its staff.



Case Study: Netflix



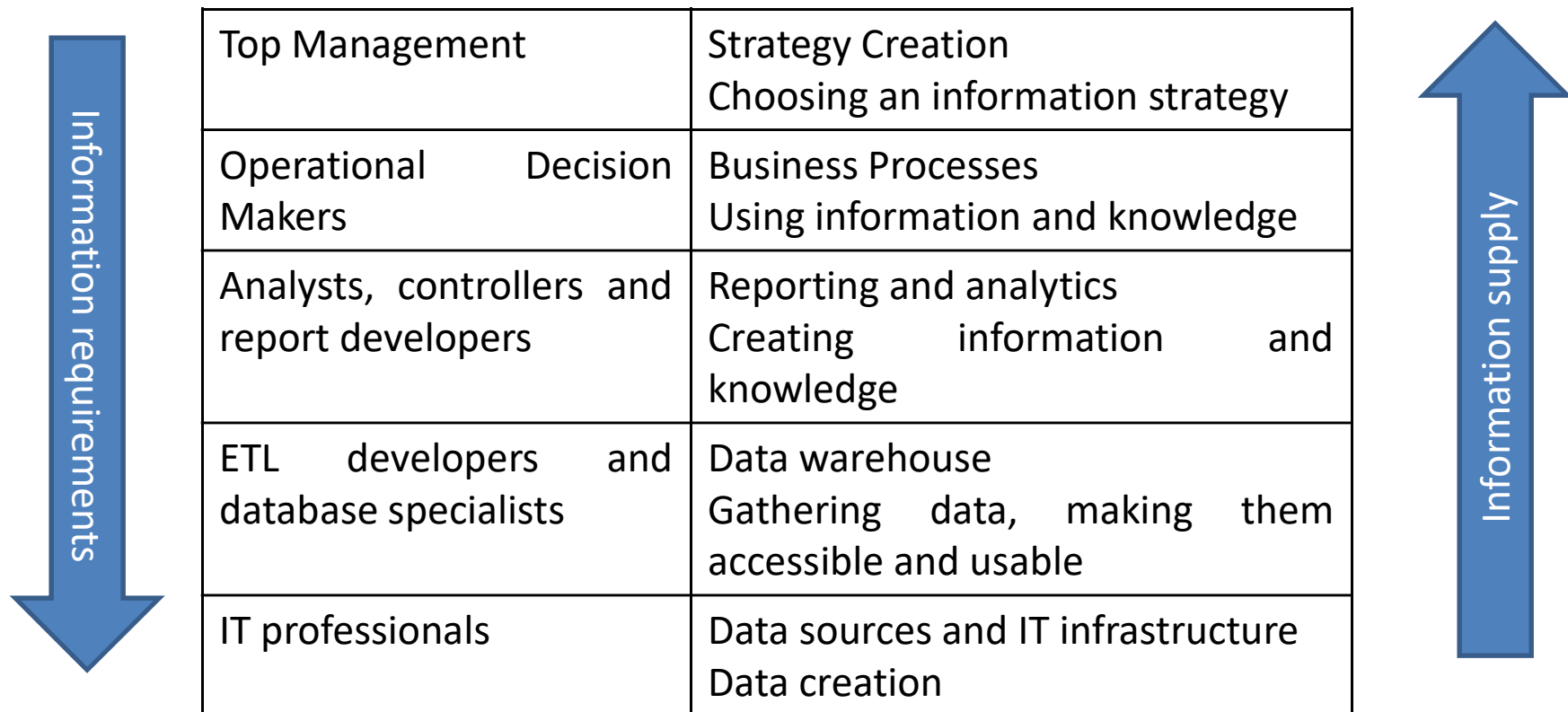
- Personalized recommendations.
- Inventory Management.
- Price to pay for the rights to distribute new DVDs.

Advanced Analytics Approach

- It provides an outline for understanding and creating successful business analytics in any type of organization.
- The purpose of the model is to give the organization a single common frame of reference for an overall structure in the creation of successful BA.
- It clarifies the roles of the individual contributors and the interaction in the information generation and information consumption process.

Advanced Analytics Approach

Business-driven environment

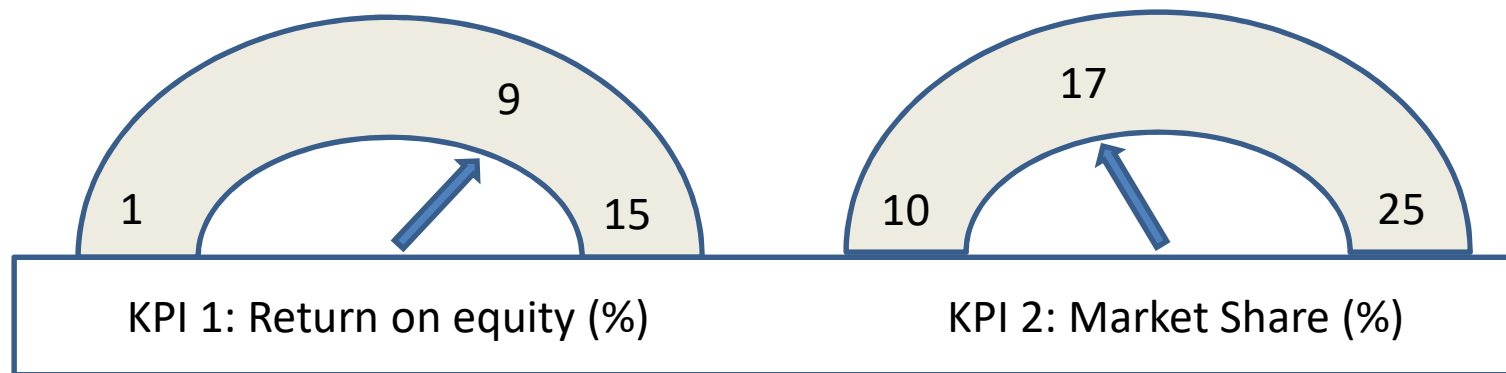


Technically Oriented Environment

Information Strategy for a Radio Station

- This case study features the Business Analytics initiative of a large radio stations that broadcasts nationwide.
- The Mission is that there is a demand for radio entertainment in the shape of good music, entertaining talk, and news.
- Vision is to become a leading player in the national market.
- The business goal is a market share of 25% and a return on equity(ROE) of 15%.

Executive Management Cockpit



Goal(KPI 1): ROE = 15%, Actual = 9%

Goal(KPI 2): Market share = 25%, Actual = 17%

Strategic Initiative by Program Manager:

“From our current record of holding on to our listeners for 15 minutes, before he or she changes channel, we must in the future hold on to our average listener for 30 minutes.”

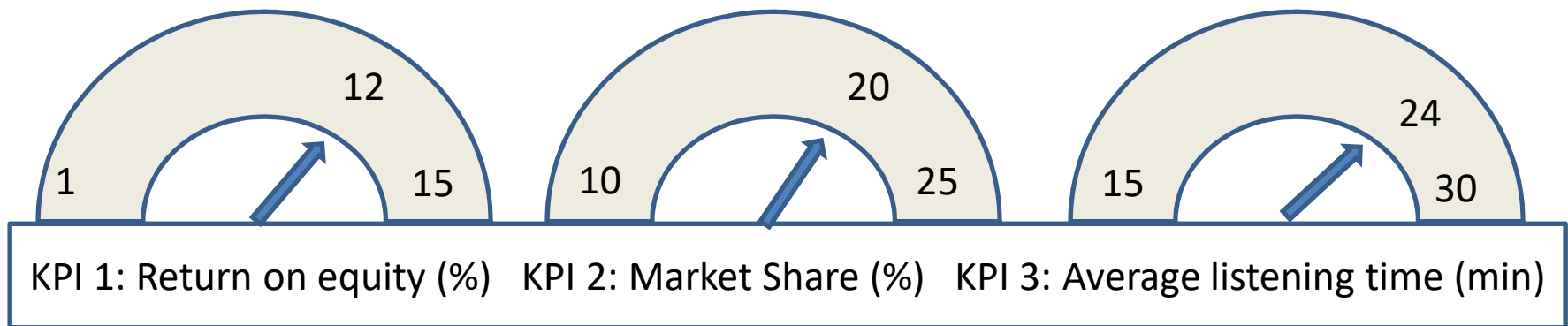
How to Make an Information Strategy for a Radio Station

- Business Process and Actions
 - Processes must be adapted to a listener profile
 - To enable the DJs and newsreaders of the radio station to continuously deliver content that is to the current listener's taste.
- Analytical Processes and Front-ends
 - Who are our listeners?
 - What do they like to listen to?
 - Who listens to what and when?
- Data Warehouse
 - Data on listeners ages, genders, and tastes and preferences 24/7

How to Make an Information Strategy for a Radio Station

- Data Sources: IT Operations and Development
 - Questionnaire to collect the data on the radio station's web site
 - Announce prizes
 - Typical listener in the early hours of the morning is a fun-loving 30 year old woman.
- Evaluation of BA process
 - In 6 months, the radio station succeeds in holding on to its average listener for 9 minutes longer than before and all three KPI are improved
 - KPI 1 = 12%, KP2 = 20% KP3=24 min

Evaluation of Strategic Initiative



Of course, this is what we have always been doing or tried to do, but it's the first time I am able to put it into words and see our endeavors in a useful analytical model.

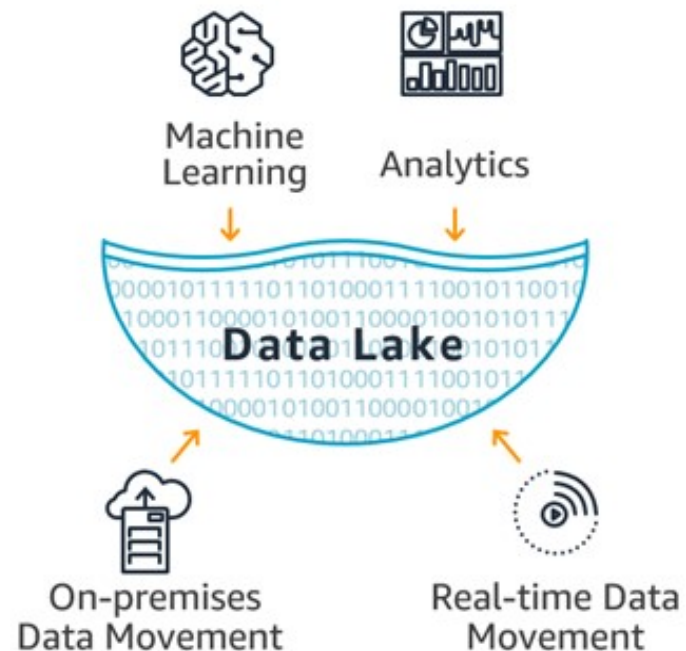
- Program Manager

Types of Data

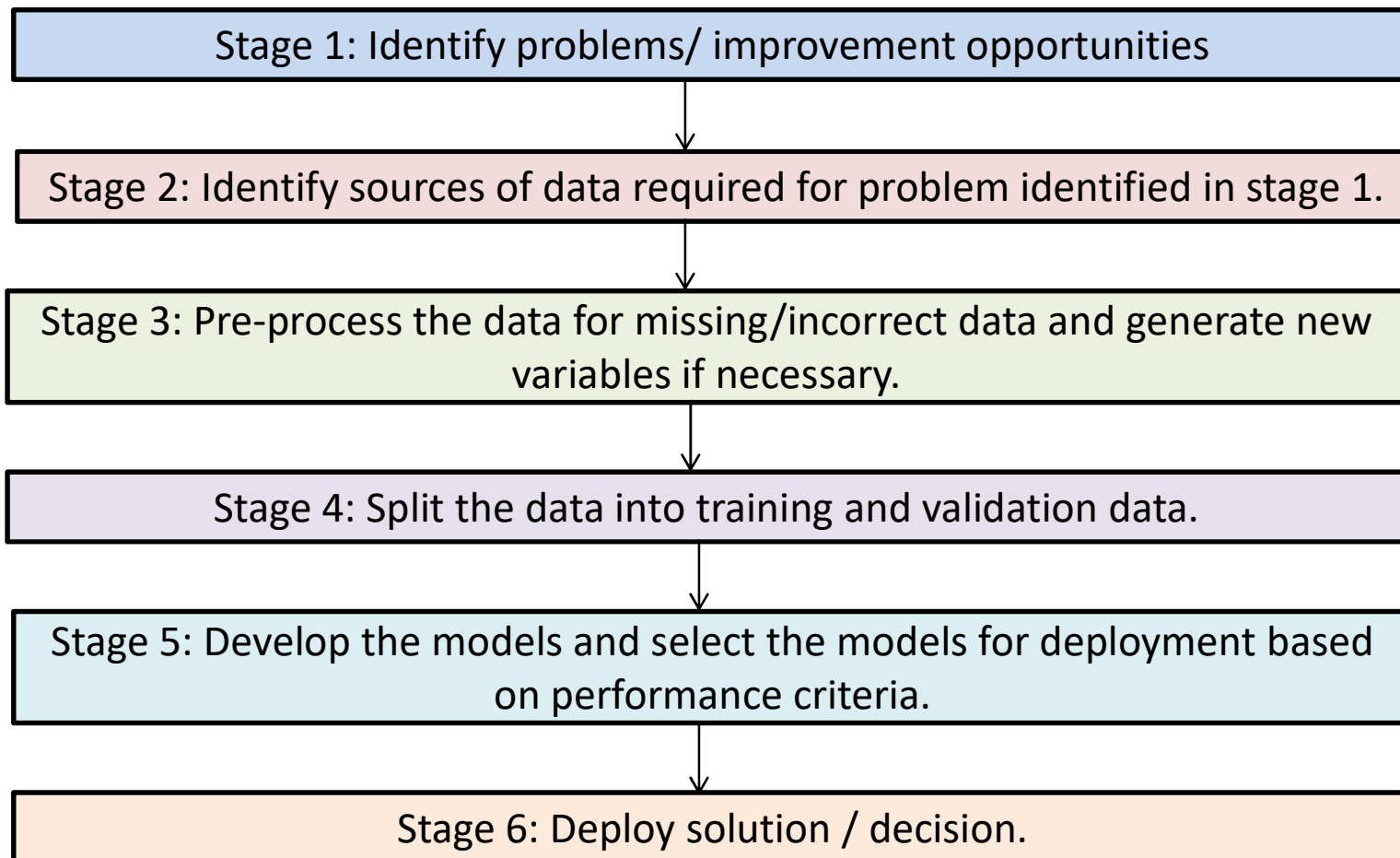
- Big Data includes huge volume, high velocity, and extensible variety of data.
- These are 3 types:
 - Structured data is numeric or alphanumeric data that could be organized in the matrix form with rows and columns. E.g. Relational Databases
 - Semi-structured data have some organizational properties that make it easier to analyse. E.g. NoSQL databases
 - Unstructured data is a data which is not organized in a predefined manner or does not have a predefined data model. E.g. text, image, audio, video

Data Used

- A data lake is a data repository that stores structured, semi-structured and unstructured data.
- All the data irrespective of its type is physically stored in files.
- This raw data is extracted, loaded and transformed for visualization, processing and analytics to be performed.
- The data lake could be created in the organization's data centre or in the cloud.



Data Driven Decision Making



Scope of Advanced Analytics



Data Measurement Scales

- The measurement scale is important in model specification for building analytics model such as regression.
- Types of Scales
 - Nominal Scale : Data placed in categories according to a specified characteristic e.g. customer location, employee designation
 - Ordinal Scale: Data that is ranked or ordered according to some relationship with one another. e.g. ranks.
 - Interval Scale: Ordinal data but with constant differences between observations. e.g. temperature
 - Ratio Scale: Continuous values and have a natural zero point. e.g. monthly sales

Classifying Data Elements in a Purchasing Database

	A	B	C	D	E	F	G	H
1	Purchase Orders							
2								
3	Supplier	Order No	Item No.	Item Description	Item Cost	Quantity	Cost per order	A/P Terms (Mo
4	Spacetime Technologies	A0111	6489	O-Ring	\$ 3.00	900	\$ 2,700.00	25
5	Steelpin Inc.	A0115	5319	Shielded Cable/ft.	\$ 1.10	17,500	\$ 19,250.00	30
6	Steelpin Inc.	A0123	4312	Bolt-nut package	\$ 3.75	4,250	\$ 15,937.50	30
7	Steelpin Inc.	A0204	5319	Shielded Cable/ft.	\$ 1.10	16,500	\$ 18,150.00	30
8	Steelpin Inc.	A0205	5677	Side Panel	\$195.00	120	\$ 23,400.00	30
9	Steelpin Inc.	A0207	4312	Bolt-nut package	\$ 3.75	4,200	\$ 15,750.00	30
10	Alum Sheeting	A0223	4224	Bolt-nut package	\$ 3.95	4,500	\$ 17,775.00	30
11	Alum Sheeting	A0433	5417	Control Panel	\$255.00	500	\$ 127,500.00	30
12	Alum Sheeting	A0443	1243	Airframe fasteners	\$ 4.25	10,000	\$ 42,500.00	30

Classifying Data Elements in a Purchasing Database

	A	B	C	D	E	F	G	H
1	Purchase Orders							
2								
3	Supplier	Order No	Item No.	Item Description	Item Cost	Quantity	Cost per order	A/P Terms (Mo
4	Spacetime Technologies	A0111	6489	O-Ring	\$ 3.00	900	\$ 2,700.00	25
5	Steelpin Inc.	A0115	5319	Shielded Cable/ft.	\$ 1.10	17,500	\$ 19,250.00	30
6	Steelpin Inc.	A0123	4312	Bolt-nut package	\$ 3.75	4,250	\$ 15,937.50	30
7	Steelpin Inc.	A0204	5319	Shielded Cable/ft.	\$ 1.10	16,500	\$ 18,150.00	30
8	Steelpin Inc.	A0205	5677	Side Panel	\$195.00	120	\$ 23,400.00	30
9	Steelpin Inc.	A0207	4312	Bolt-nut package	\$ 3.75	4,200	\$ 15,750.00	30
10	Alum Sheeting	A0223	4224	Bolt-nut package	\$ 3.95	4,500	\$ 17,775.00	30
11	Alum Sheeting	A0433	5417	Control Panel	\$255.00	500	\$ 127,500.00	30
12	Alum Sheeting	A0443	1243	Airframe fasteners	\$ 4.25	10,000	\$ 42,500.00	30

Categorical
Categorical
Categorical
Categorical
Ratio
Ratio
Ratio
Ratio
Interval
Interval

Descriptive Analytics

The use of data to understand past and current business performance and make informed decisions. Provides reports and information on what happened in a past state.

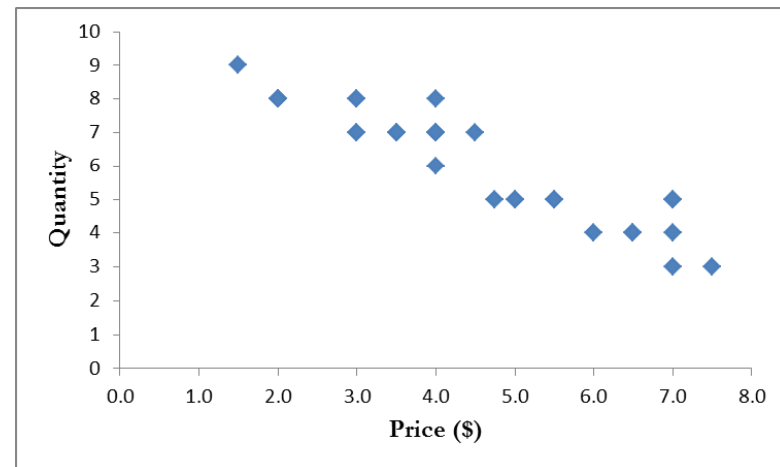
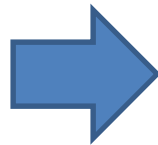
- How much did we sell in each region?
- What was our revenue and profit last quarter?
- How many and what types of complaints did we resolve?
- Which factory has the lowest productivity?



Data Visualization, Descriptive Measures, Probability Distribution

Example: Price Demand Model

Price (\$)	Demand
4.0	7
3.5	7
5.0	5
...	...
2.0	8



Predictive Analytics

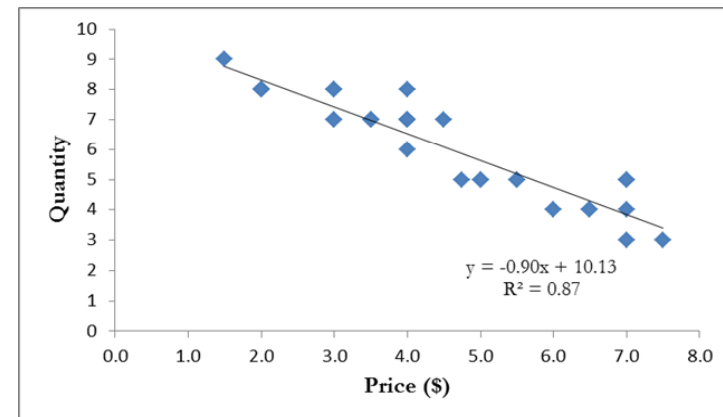
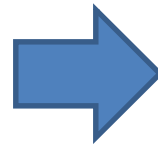
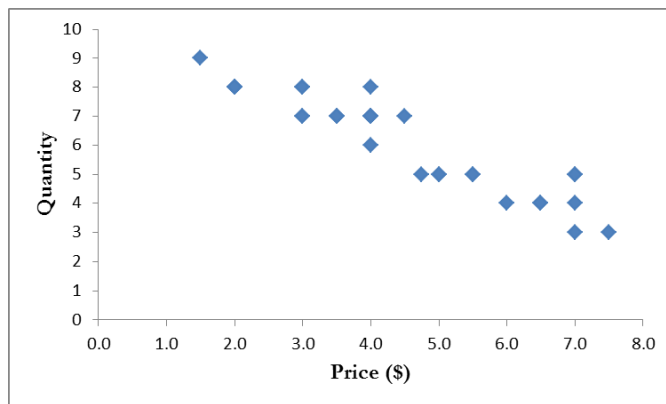
It analyzes past performance in an effort to predict the future by examining historical data, detecting patterns or relationships in these data, and then extrapolating these relationships forward in time. Attempts to provide information on when and what event will happen in a future state.

- What will happen if demand falls by 10% or if supplier prices go up by 5%?
- What is the risk of losing money in a new business venture?

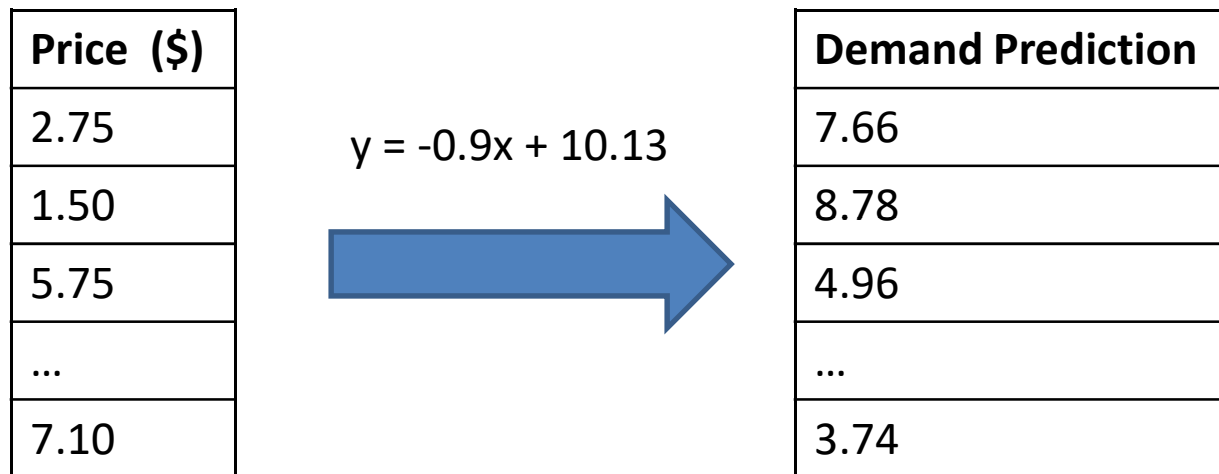


Predictive Modeling, Regression Analysis, Simulation, Forecasting

Example: Price Demand Model



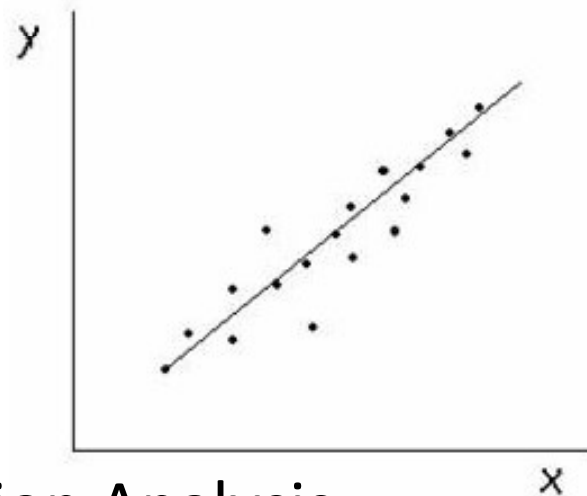
Demand Prediction



Prescriptive Analytics

It uses optimization to identify the best alternative to minimize or maximize some objective. Attempts to answer what, when and why something is going to happen and then recommends action.

- How much should we produce to maximize profit?
- Should we change our plan if a natural disaster closes a supplier's factory: if so, by how much?

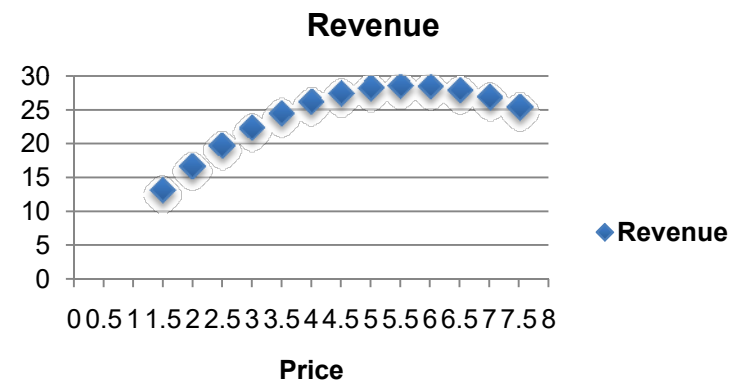
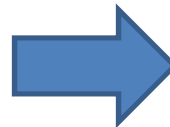
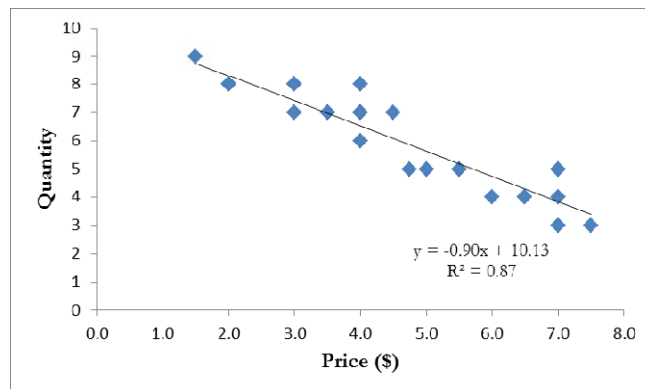


Optimization techniques, Decision Analysis

Example: Price Demand Model

Goal 1: Maximize the quantity sold

Goal 2: Maximize revenue



Python Libraries

Numpy
Mathematical Computation
www.numpy.org

Scipy
Statistical Computation
www.scipy.org

Pandas
Data Structure Operations
pandas.pydata.org

StatsModelling
Statistical Modelling
www.statsmodels.org

Matplotlib
Visualization
[Matplotlib.org](http://matplotlib.org)

Sklearn

Advanced Analytics Applications

Banks	predict and prevent credit frauds
Manufacturers	production planning, purchasing and inventory management
Retailers	predict the best location for stores and how to stock them.
Sports	game strategy and optimal ticket prices
Consumer and Industrial Goods	optimal pricing
Insurer and credit card firms	target consumer segments.

Summary

- A new analytical ecosystem that supports the entire spectrum of traditional and big data analytical applications.
- Successful Advanced Analytics initiatives are always closely interlinked with the organization's strategy (mission, vision, and goals) and are put in the right direction toward business objective.
- Advanced Analytics is not purely an IT discipline.
 - Advanced Analytics initiatives floating aimlessly
 - Driven by the organization's technical environment

References

- Business Analytics for Managers, Laursen and Thorlund, Wiley Publication
- James Evans, Business Analytics: Methods, Models and Decisions, Second Edition, Pearson Publication, 2017.
- Manaranjan Pradhan and U Dinesh Kumar, Machine Learning using Python, Wiley Publication, 2019.
- U Dinesh Kumar, Business Analytics- The Science of Data-Driven Decision Making, Wiley Publication, 2017.

*Thank
You*